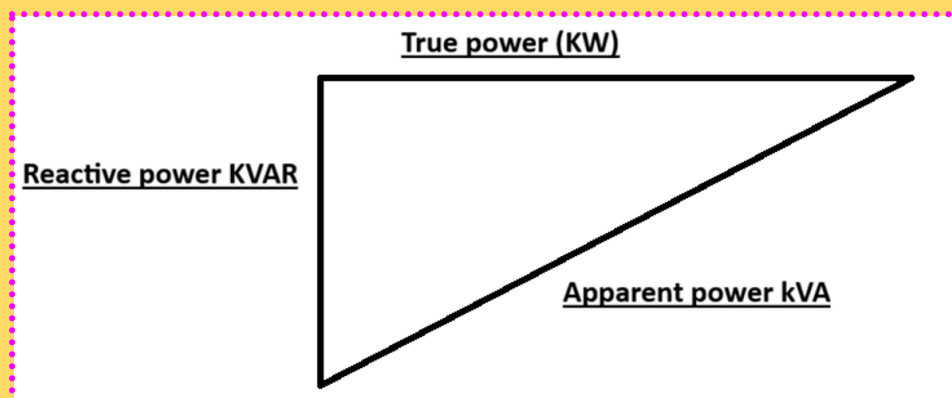


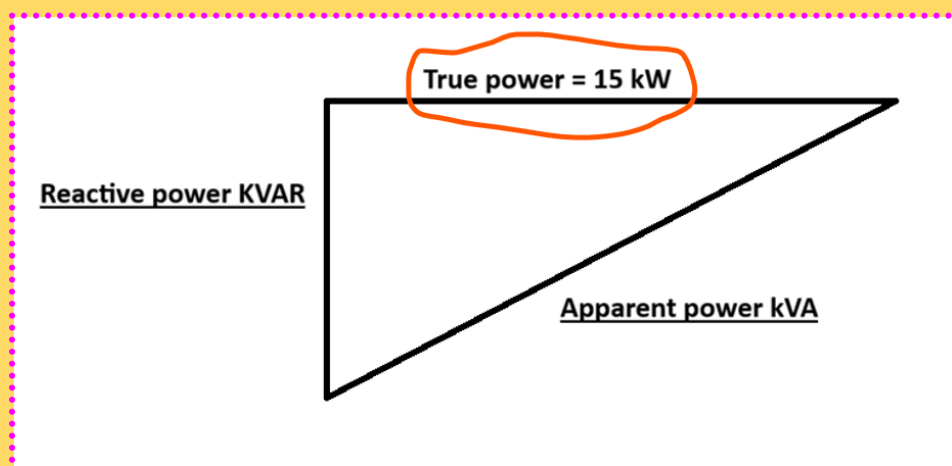
Unit 302 Mock exam

Question 9: Draw a power triangle to scale for a 15 kW machine with a p.f of 0.74?

Step one - draw base formula

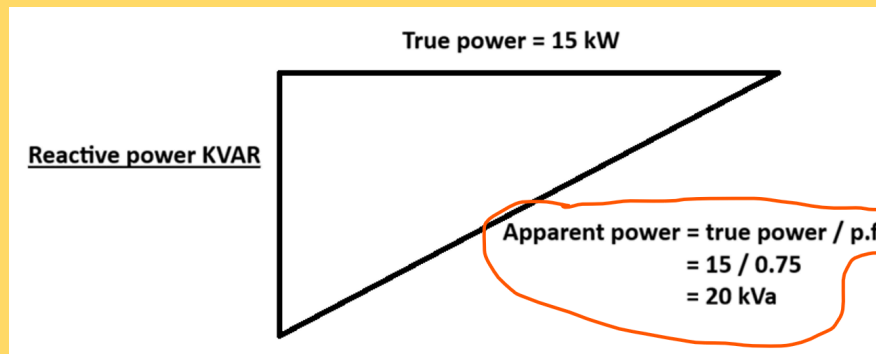


Swap in 15 kW as the true power. This was provided in the question.

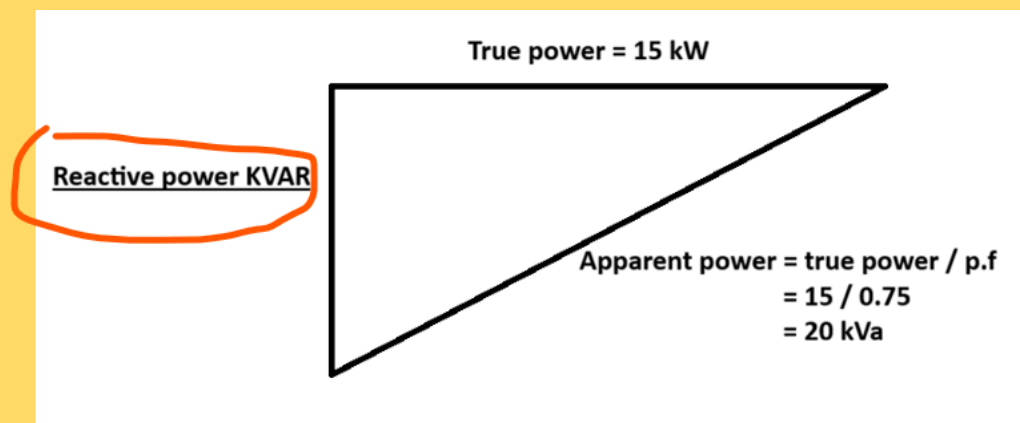


Step two - calculate the apparent power

$$\text{Apparent power} = \text{true power} / \text{p.f}$$



Step 3 - Calculate the Reactive power KVAR

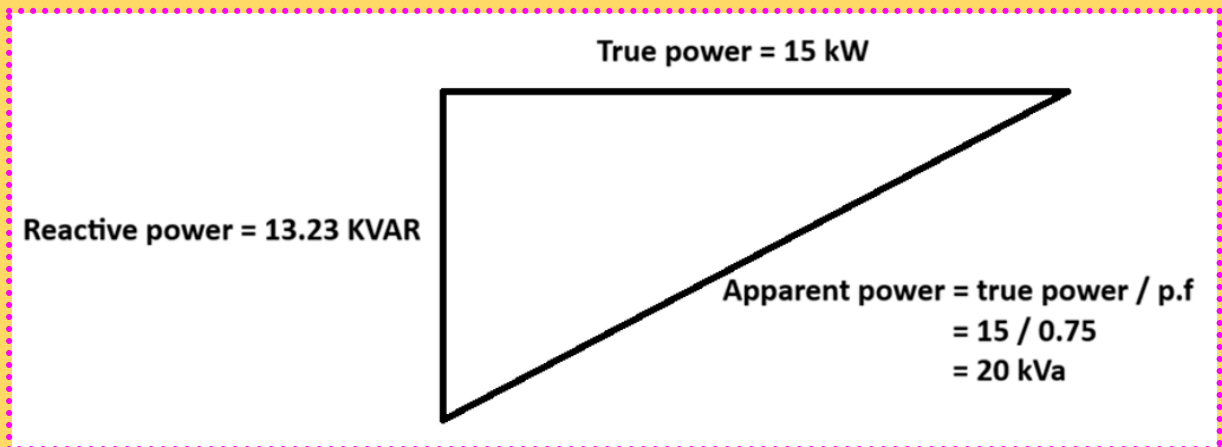


$$\text{Reactive power} = \sqrt{\text{Apparent power}^2 - \text{True power}^2}$$

Apparent power is first
True power is second

$$\begin{aligned} & \sqrt{20^2 - 15^2} \\ &= \sqrt{400 - 225} \\ &= \sqrt{175} \approx 13.23 \end{aligned}$$

Update power triangle with information.



Step 4 - Calculate the power factor

$$\cos^{-1}(0.75) = 41.41$$

